



Multi-Sample Comparison Report

Hydraulic Conductivity Analysis - 5 Samples

Project: Project
Date: June 30, 2026

Multi-Sample Comparison Report

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Report Date:	June 30, 2026 at 09:48

Number of Samples:	5
Temperature:	20 °C
Porosity:	Varies by sample (0.26 to 0.407)

Executive Summary

Comparison Analysis: This report compares 5 soil samples based on grain size distribution and hydraulic conductivity estimates. Key Findings: - Highest permeability: Sample_1_Odong (9.54e-04 m/s) - Lowest permeability: Example_2_Case_1_Vukovic_15C (1.14e-06 m/s)
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- Permeability range: 837.4x difference

Methodology

Comparative Analysis Approach

This comparison report presents a side-by-side analysis of multiple soil samples to identify patterns, variations, and relationships between grain size characteristics and hydraulic conductivity estimates.

Analysis Components

Grain Size Comparison: Overlapping distribution curves allow visual assessment of particle size variations between samples.

K-Value Comparison: Box plots and statistical summaries reveal the range and reliability of hydraulic conductivity estimates across samples.

Method Reliability: A reliability matrix shows which empirical methods are applicable for each sample, helping identify the most suitable estimation approaches.

Results & Analysis

Sample Overview

Sample	Temp (°C)	Porosity	D10 (mm)	D50 (mm)	D60 (mm)	Cu
Example_1_Case_1_Vukovic	20.00	0.384	0.150	0.485	0.550	3.67
Example_1_Case_2_Vukovic_15C	20.00	0.384	0.150	0.480	0.550	3.67
Example_2_Case_1_Vukovic_15C	20.00	0.26	0.014	0.210	0.300	21.43
Sample_1_Odong	20.00	0.346	0.341	1.181	1.886	5.53
Thomson_SERDP_Borden_Sand	20.00	0.407	0.070	0.180	0.195	2.79

Sample	Soil type	Descriptor	K geometric mean (m/s)
Example_1_Case_1_Vukovic	Poorly-graded sand	Moderately well sorted sand low in fines	2.73e-04
Example_1_Case_2_Vukovic_15C	Poorly-graded sand	Moderately well sorted sand low in fines	2.74e-04
Example_2_Case_1_Vukovic_15C	Well-graded sand	Poorly sorted silty gravelly sand with fines	1.14e-06
Sample_1_Odong	Poorly-graded sand	Poorly sorted gravelly sand low in fines	9.54e-04
Thomson_SERDP_Borden_Sand	Poorly-graded sand	Moderately well sorted sand with fines	5.01e-05

Grain Parameters Comparison

Parameter	Example_1_Case_1_Vukovic	Example_1_Case_2_Vukovic_15C	Example_2_Case_1_Vukovic_15C	Sample_1_Odong	Thomson_SERDP_Borden_Sand	Statistics
D10 (mm)	0.150	0.150	0.014	0.341	0.070	Mean: 0.145 Std: 0.111
D50 (mm)	0.485	0.480	0.210	1.181	0.180	Mean: 0.507 Std: 0.361
D60 (mm)	0.550	0.550	0.300	1.886	0.195	Mean: 0.696 Std: 0.611
Cu	3.67	3.67	21.43	5.53	2.79	Mean: 7.42 Std: 7.06

Parameter	Example_1_Case_1_Vukovic	Example_1_Case_2_Vukovic_15C	Example_2_Case_1_Vukovic_15C	Sample_1_Odong	Thomson_SERDP_Borden_Sand	Statistics
Cc	1.48	1.57	1.50	0.54	1.56	Mean: 1.33 Std: 0.40

K-Value Aggregate Summary

Scope	Datasets	Geometric mean (m/s)	Arithmetic mean (m/s)	Median (m/s)	ln(K) std dev
Overall	5	1.32e-04	4.53e-04	2.48e-04	2.207
g	3	9.13e-05	2.84e-04	2.48e-04	2.454
g2	2	2.33e-04	7.11e-04	3.49e-04	1.609

Scope	Included K cells	Warning cells	Permeability class
Overall	58 / 80	22	High (Clean Sand)
g	35 / 48	13	Moderate (Fine Sand)
g2	23 / 32	9	High (Clean Sand)

K-Value Calculations by Dataset and Method

Sample	Method	K (m/s)	Status	Included	Notes
Example_1_Case_1_Vukovic	Hazen	3.00e-04	OK	Yes	OK
Example_1_Case_1_Vukovic	Hazen_1892	2.25e-04	Error	No	Empirical mm² variant (10°C reference)
Example_1_Case_1_Vukovic	Slichter	9.60e-05	OK	Yes	OK
Example_1_Case_1_Vukovic	Terzaghi	1.67e-04	OK	Yes	Reference 10°C calibration
Example_1_Case_1_Vukovic	Beyer	2.48e-04	OK	Yes	Within recommended range
Example_1_Case_1_Vukovic	Sauerbrei	3.39e-04	OK	Yes	OK
Example_1_Case_1_Vukovic	Kruger	5.16e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
Example_1_Case_1_Vukovic	Kozeny-Carman	1.03e-03	OK	Yes	OK
Example_1_Case_1_Vukovic	Zunker	5.64e-04	OK	Yes	OK
Example_1_Case_1_Vukovic	Zamarin	6.65e-04	OK	Yes	OK
Example_1_Case_1_Vukovic	USBR	2.55e-04	OK	Yes	OK
Example_1_Case_1_Vukovic	Barr	1.34e-04	OK	Yes	OK
Example_1_Case_1_Vukovic	Alyamani-Sen	8.38e-05	OK	Yes	OK
Example_1_Case_1_Vukovic	Chapuis	1.74e-04	OK	Yes	OK
Example_1_Case_1_Vukovic	Shepherd	5.01e-04	OK	Yes	OK

Sample	Method	K (m/s)	Status	Included	Notes
Example_1_Case_1_Vukovic	Krumbein-Monk	3.22e-04	OK	Yes	OK
Example_1_Case_2_Vukovic_15C	Hazen	3.00e-04	OK	Yes	OK
Example_1_Case_2_Vukovic_15C	Hazen_1892	2.25e-04	Error	No	Empirical mm ² variant (10°C reference)
Example_1_Case_2_Vukovic_15C	Slichter	9.60e-05	OK	Yes	OK
Example_1_Case_2_Vukovic_15C	Terzaghi	1.67e-04	OK	Yes	Reference 10°C calibration
Example_1_Case_2_Vukovic_15C	Beyer	2.48e-04	OK	Yes	Within recommended range
Example_1_Case_2_Vukovic_15C	Sauerbrei	3.39e-04	OK	Yes	OK
Example_1_Case_2_Vukovic_15C	Kruger	4.36e-04	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
Example_1_Case_2_Vukovic_15C	Kozeny-Carman	1.13e-03	OK	Yes	OK
Example_1_Case_2_Vukovic_15C	Zunker	5.65e-04	OK	Yes	OK
Example_1_Case_2_Vukovic_15C	Zamarin	6.08e-04	OK	Yes	OK
Example_1_Case_2_Vukovic_15C	USBR	2.55e-04	OK	Yes	OK
Example_1_Case_2_Vukovic_15C	Barr	1.34e-04	OK	Yes	OK
Example_1_Case_2_Vukovic_15C	Alyamani-Sen	8.63e-05	OK	Yes	OK
Example_1_Case_2_Vukovic_15C	Chapuis	1.74e-04	OK	Yes	OK
Example_1_Case_2_Vukovic_15C	Shepherd	4.92e-04	OK	Yes	OK
Example_1_Case_2_Vukovic_15C	Krumbein-Monk	3.35e-04	OK	Yes	OK
Example_2_Case_1_Vukovic_15C	Hazen	1.16e-06	Warning	No	Best for D10: 0.01-0.3 cm, UC < 5
Example_2_Case_1_Vukovic_15C	Hazen_1892	1.96e-06	Warning	No	D10 outside 0.01–0.3 cm; Empirical mm ² variant (10°C reference)
Example_2_Case_1_Vukovic_15C	Slichter	2.32e-07	OK	Yes	OK
Example_2_Case_1_Vukovic_15C	Terzaghi	3.36e-07	OK	Yes	Reference 10°C calibration
Example_2_Case_1_Vukovic_15C	Beyer	1.38e-06	Warning	No	Outside stated U or D10 limits
Example_2_Case_1_Vukovic_15C	Sauerbrei	1.24e-06	OK	Yes	OK
Example_2_Case_1_Vukovic_15C	Kruger	1.14e-06	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
Example_2_Case_1_Vukovic_15C	Kozeny-Carman	8.17e-07	OK	Yes	OK

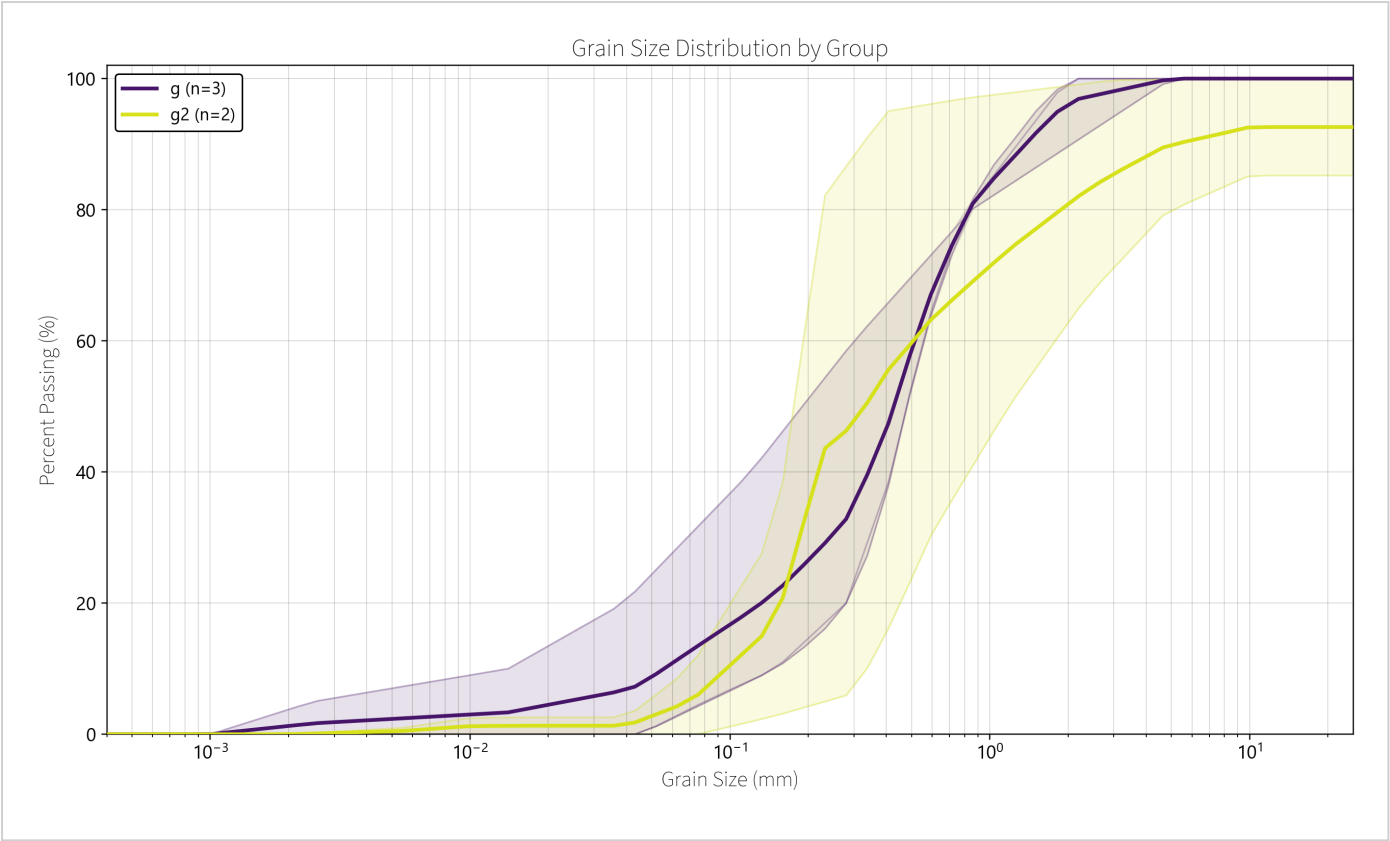
Sample	Method	K (m/s)	Status	Included	Notes
Example_2_Case_1_Vukovic_15C	Zunker	7.16e-07	Warning	No	Material contains fractions finer than 0.0025 mm
Example_2_Case_1_Vukovic_15C	Zamarin	1.00e-06	OK	Yes	OK
Example_2_Case_1_Vukovic_15C	USBR	2.74e-06	Warning	No	$U \geq 5$ exceeds USBR limit
Example_2_Case_1_Vukovic_15C	Barr	2.50e-07	OK	Yes	OK
Example_2_Case_1_Vukovic_15C	Alyamani-Sen	1.36e-05	Warning	No	Effective diameter non-positive
Example_2_Case_1_Vukovic_15C	Chapuis	4.13e-08	Warning	No	0.3 < n < 0.7 not satisfied; D10 > 0.10 mm not satisfied; Uniformity coefficient outside 2–12
Example_2_Case_1_Vukovic_15C	Shepherd	1.26e-04	OK	Yes	OK
Example_2_Case_1_Vukovic_15C	Krumbein-Monk	3.52e-06	Warning	No	sigma_phi = 3.07 suggests non-lognormal distribution
Sample_1_Odong	Hazen	1.29e-03	Warning	No	Best for D10: 0.01-0.3 cm, UC < 5
Sample_1_Odong	Hazen_1892	1.16e-03	Error	No	Empirical mm ² variant (10°C reference)
Sample_1_Odong	Slichter	3.54e-04	OK	Yes	OK
Sample_1_Odong	Terzaghi	6.02e-04	OK	Yes	Reference 10°C calibration
Sample_1_Odong	Beyer	1.18e-03	OK	Yes	Within recommended range
Sample_1_Odong	Sauerbrei	6.79e-04	OK	Yes	OK
Sample_1_Odong	Kruger	1.98e-03	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
Sample_1_Odong	Kozeny-Carman	3.86e-03	OK	Yes	OK
Sample_1_Odong	Zunker	2.20e-03	OK	Yes	OK
Sample_1_Odong	Zamarin	2.56e-03	OK	Yes	OK
Sample_1_Odong	USBR	7.89e-04	Warning	No	$U \geq 5$ exceeds USBR limit
Sample_1_Odong	Barr	4.51e-04	OK	Yes	OK
Sample_1_Odong	Alyamani-Sen	3.49e-04	OK	Yes	OK
Sample_1_Odong	Chapuis	7.59e-04	OK	Yes	OK
Sample_1_Odong	Shepherd	2.17e-03	OK	Yes	OK
Sample_1_Odong	Krumbein-Monk	5.93e-04	OK	Yes	OK
Thomson_SERDP_Borden_Sand	Hazen	7.20e-05	Warning	No	Best for D10: 0.01-0.3 cm, UC < 5

Sample	Method	K (m/s)	Status	Included	Notes
Thomson_SERDP_Borden_Sand	Hazen_1892	4.90e-05	Warning	No	D10 outside 0.01–0.3 cm; Empirical mm² variant (10°C reference)
Thomson_SERDP_Borden_Sand	Slichter	2.53e-05	OK	Yes	OK
Thomson_SERDP_Borden_Sand	Terzaghi	4.43e-05	OK	Yes	Reference 10°C calibration
Thomson_SERDP_Borden_Sand	Beyer	5.70e-05	OK	Yes	Within recommended range
Thomson_SERDP_Borden_Sand	Sauerbrei	6.99e-05	OK	Yes	OK
Thomson_SERDP_Borden_Sand	Kruger	4.06e-05	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
Thomson_SERDP_Borden_Sand	Kozeny-Carman	9.96e-05	OK	Yes	OK
Thomson_SERDP_Borden_Sand	Zunker	4.98e-05	Warning	No	Material contains fractions finer than 0.0025 mm
Thomson_SERDP_Borden_Sand	Zamarin	5.44e-05	OK	Yes	OK
Thomson_SERDP_Borden_Sand	USBR	2.86e-05	OK	Yes	OK
Thomson_SERDP_Borden_Sand	Barr	3.73e-05	OK	Yes	OK
Thomson_SERDP_Borden_Sand	Alyamani-Sen	3.09e-05	OK	Yes	OK
Thomson_SERDP_Borden_Sand	Chapuis	4.57e-05	Warning	No	D10 > 0.10 mm not satisfied
Thomson_SERDP_Borden_Sand	Shepherd	9.74e-05	OK	Yes	OK
Thomson_SERDP_Borden_Sand	Krumbein-Monk	6.41e-05	OK	Yes	OK

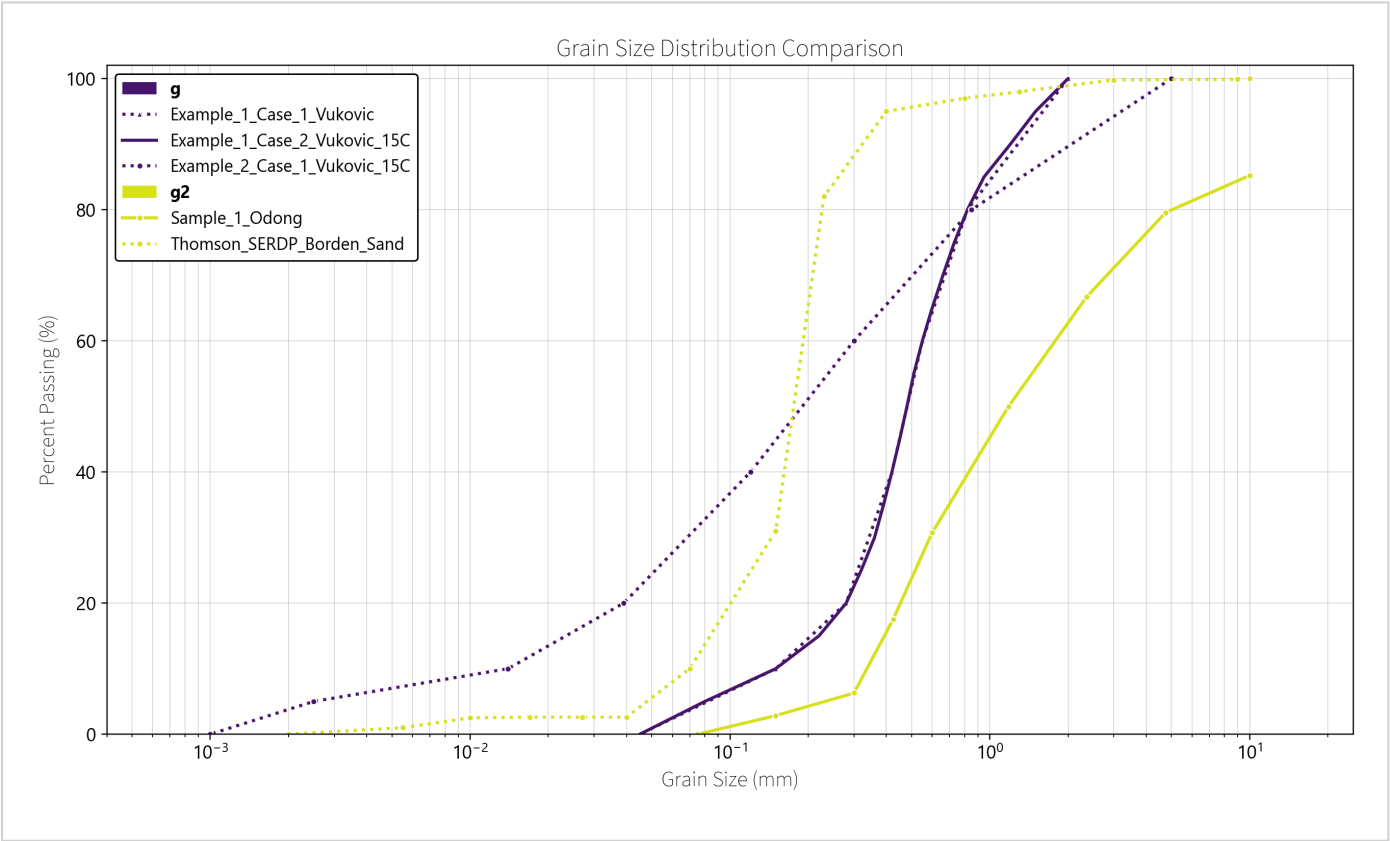
Permeability Classification Summary

Sample Name	K geometric mean (m/s)	Classification
Example_1_Case_1_Vukovic	2.73e-04	High (Clean Sand)
Example_1_Case_2_Vukovic_15C	2.74e-04	High (Clean Sand)
Example_2_Case_1_Vukovic_15C	1.14e-06	Low (Silt)
Sample_1_Odong	9.54e-04	High (Clean Sand)
Thomson_SERDP_Borden_Sand	5.01e-05	Moderate (Fine Sand)

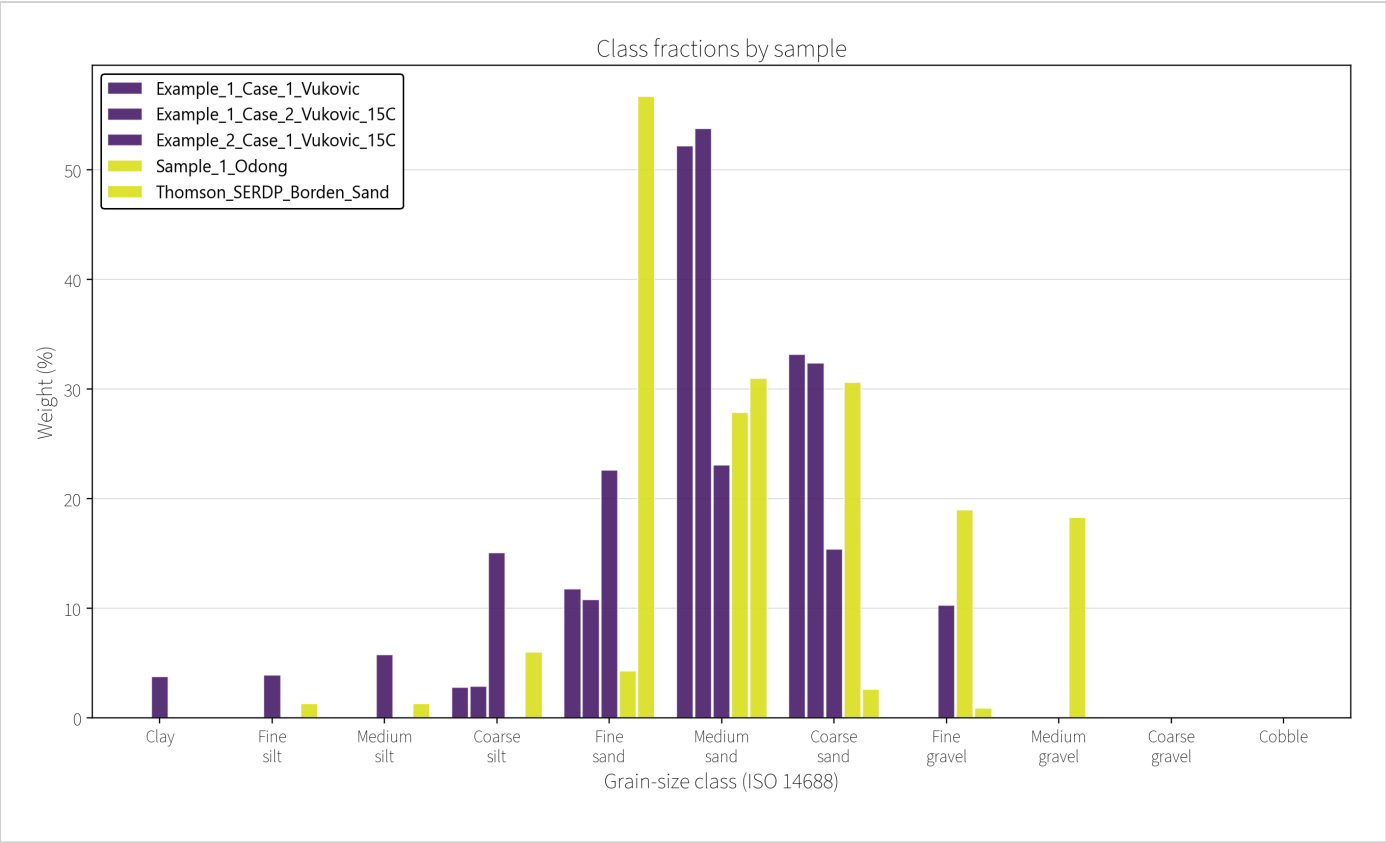
Grain Size Distribution Comparison — per group



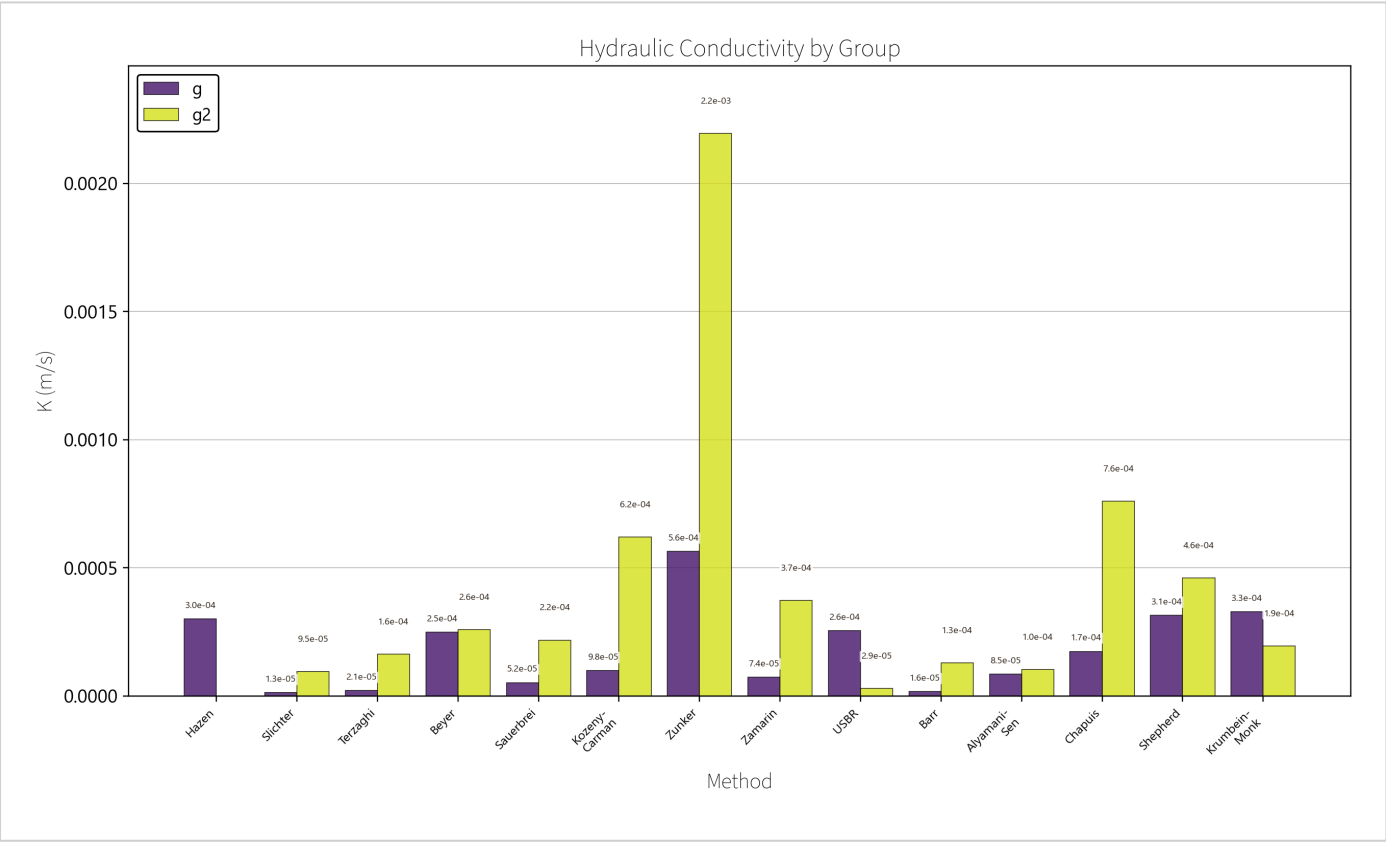
Grain Size Distribution Comparison — per dataset



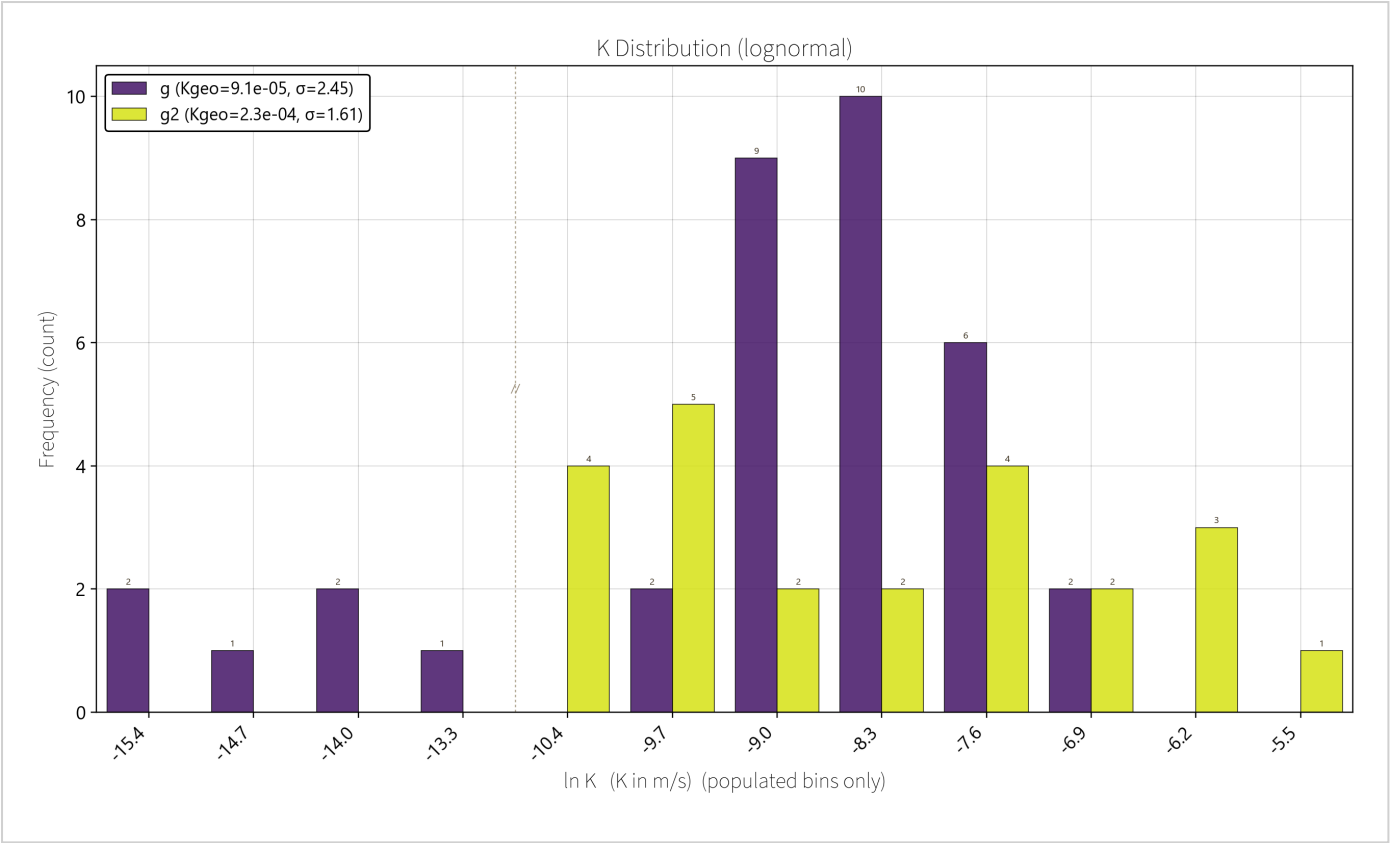
Grain-size Class Histogram



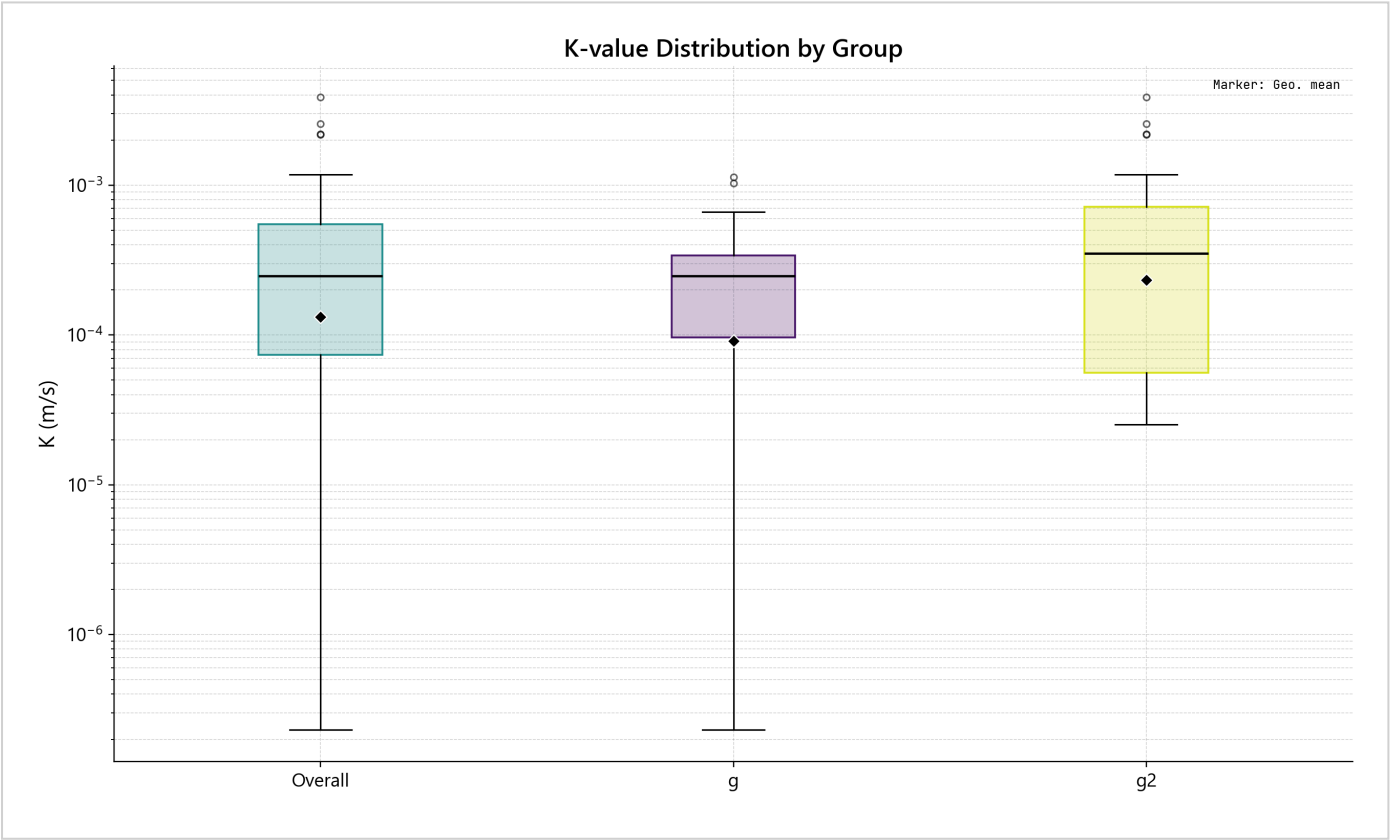
Hydraulic Conductivity by Method



Hydraulic Conductivity Distribution (Lognormal)



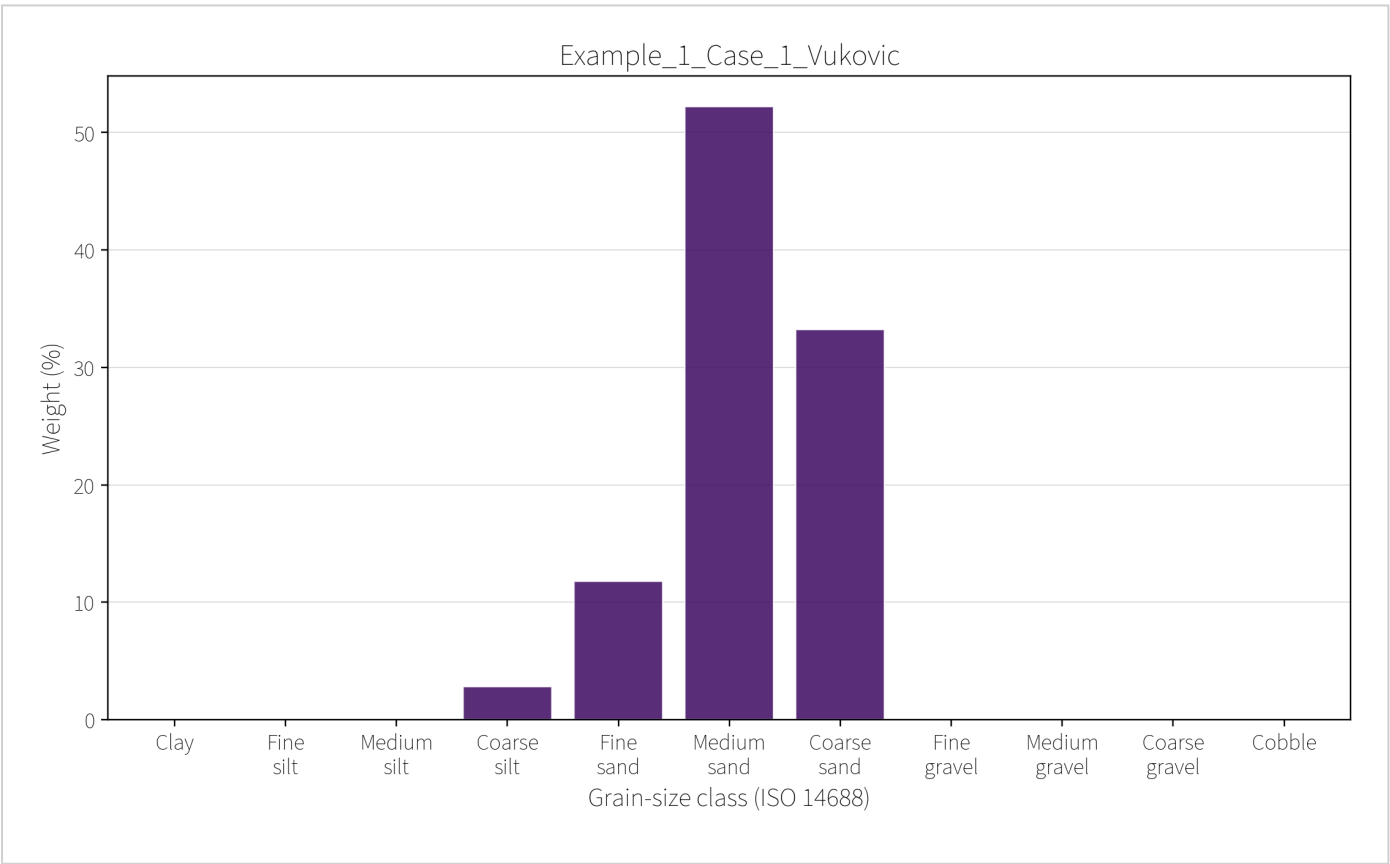
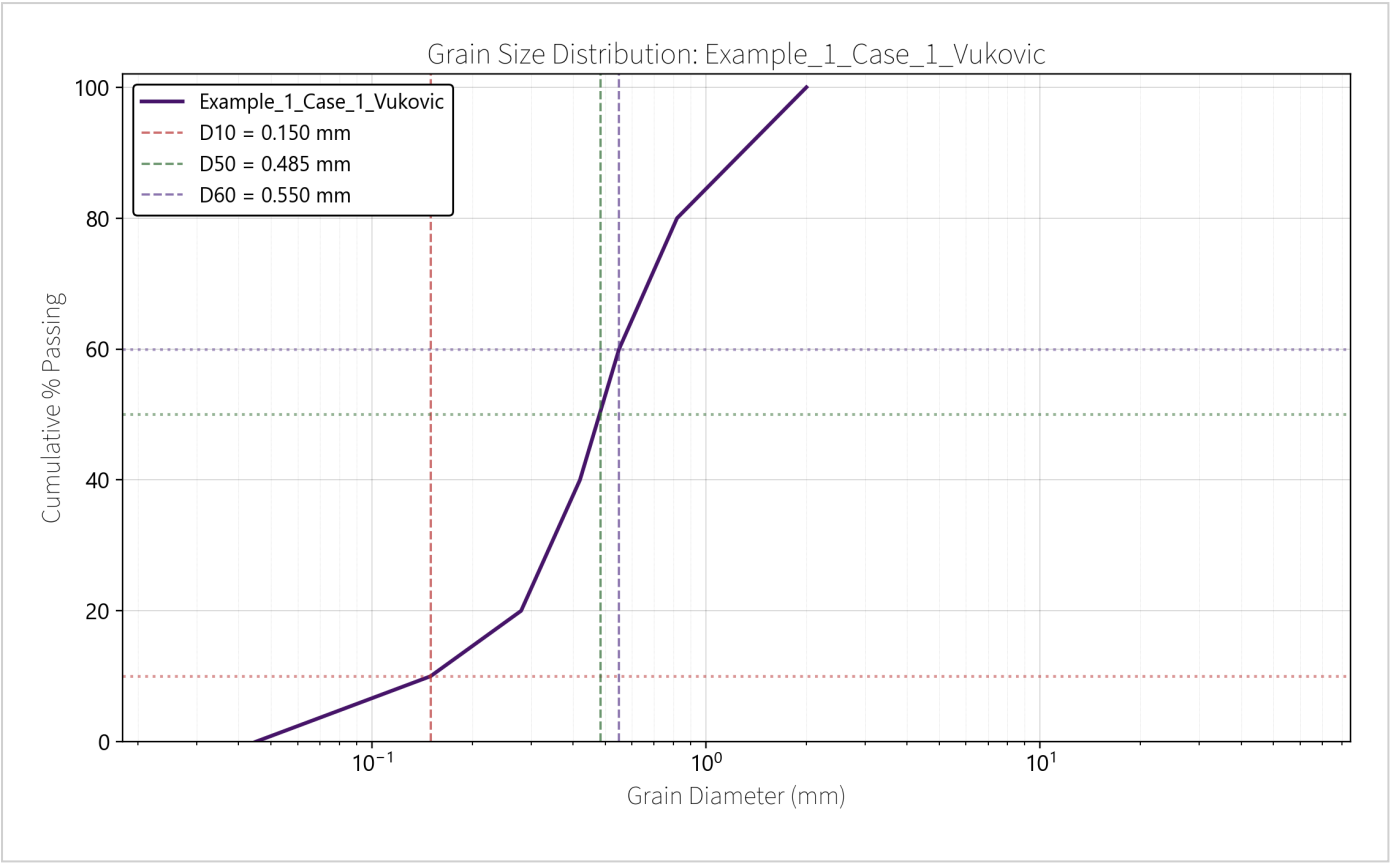
Hydraulic Conductivity Distribution



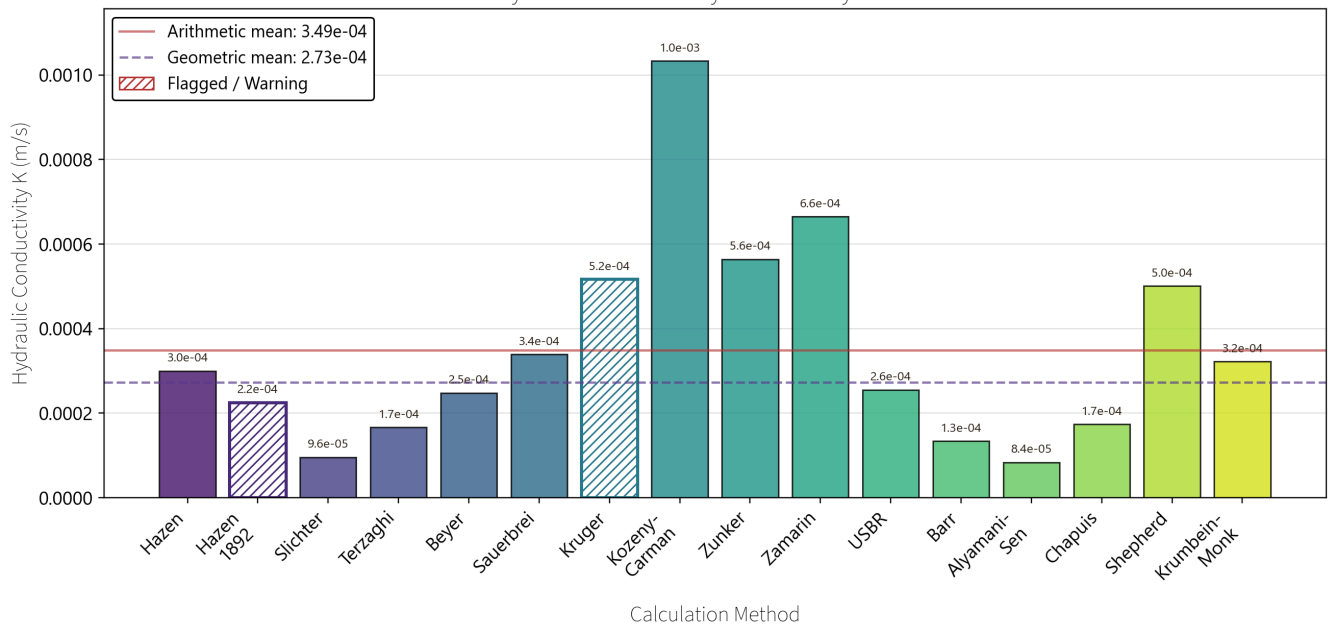
Appendix: Method Reliability Matrix

Method Applicability Matrix — All Samples					
Alyamani-Sen	OK	OK	WARN	OK	OK
Barr	OK	OK	OK	OK	OK
Beyer	OK	OK	WARN	OK	OK
Chapuis	OK	OK	WARN	OK	WARN
Hazen	OK	OK	WARN	WARN	WARN
Hazen_1892	ERR	ERR	WARN	ERR	WARN
Kozeny-Carman	OK	OK	OK	OK	OK
Kruger	WARN	WARN	WARN	WARN	WARN
Krumbein-Monk	OK	OK	WARN	OK	OK
Sauerbrei	OK	OK	OK	OK	OK
Shepherd	OK	OK	OK	OK	OK
Slichter	OK	OK	OK	OK	OK
Terzaghi	OK	OK	OK	OK	OK
USBR	OK	OK	WARN	WARN	OK
Zamarin	OK	OK	OK	OK	OK
Zunker	OK	OK	WARN	OK	WARN
Example_1_Case_1_Vukovic Example_1_Case_2_Vukovic_15C Example_2_Case_1_Vukovic_15C Sample_1_Odong Thomson_SERDP_Borden_Sand					

Example_1_Case_1_Vukovic

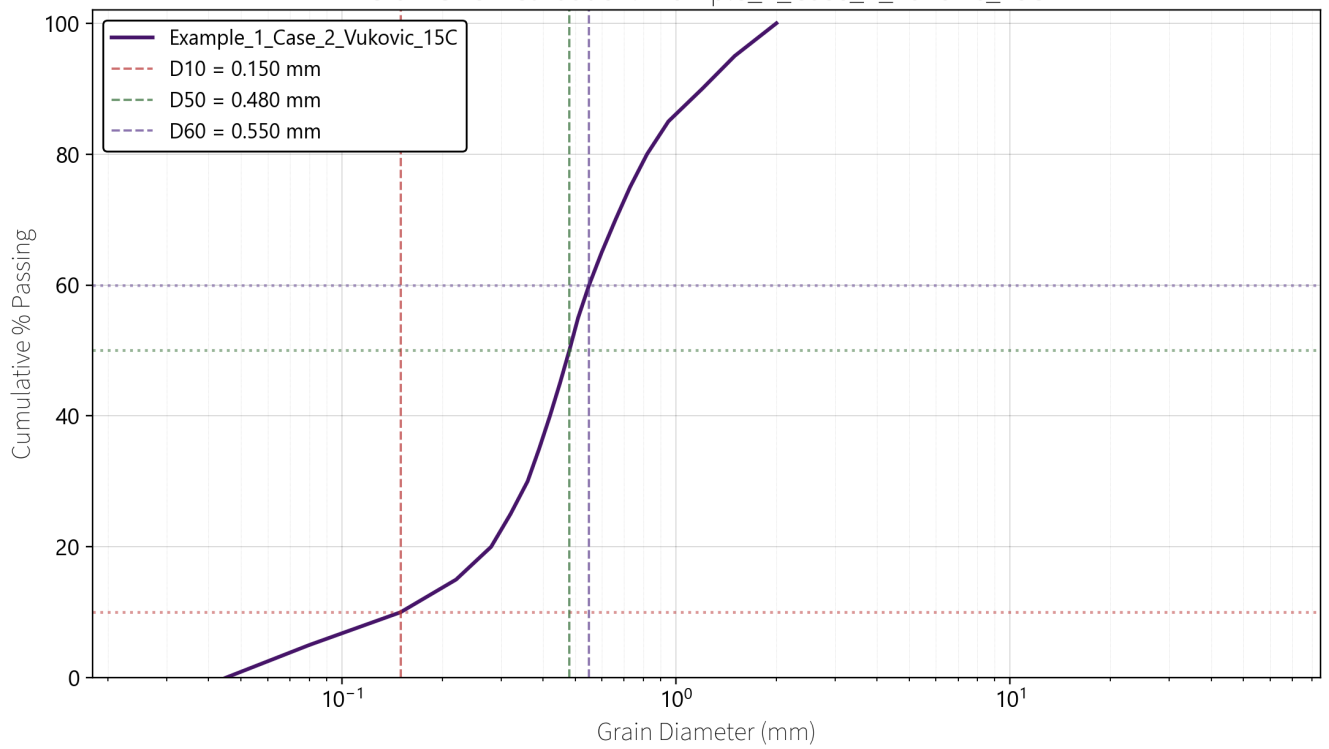


Hydraulic Conductivity Estimates by Method

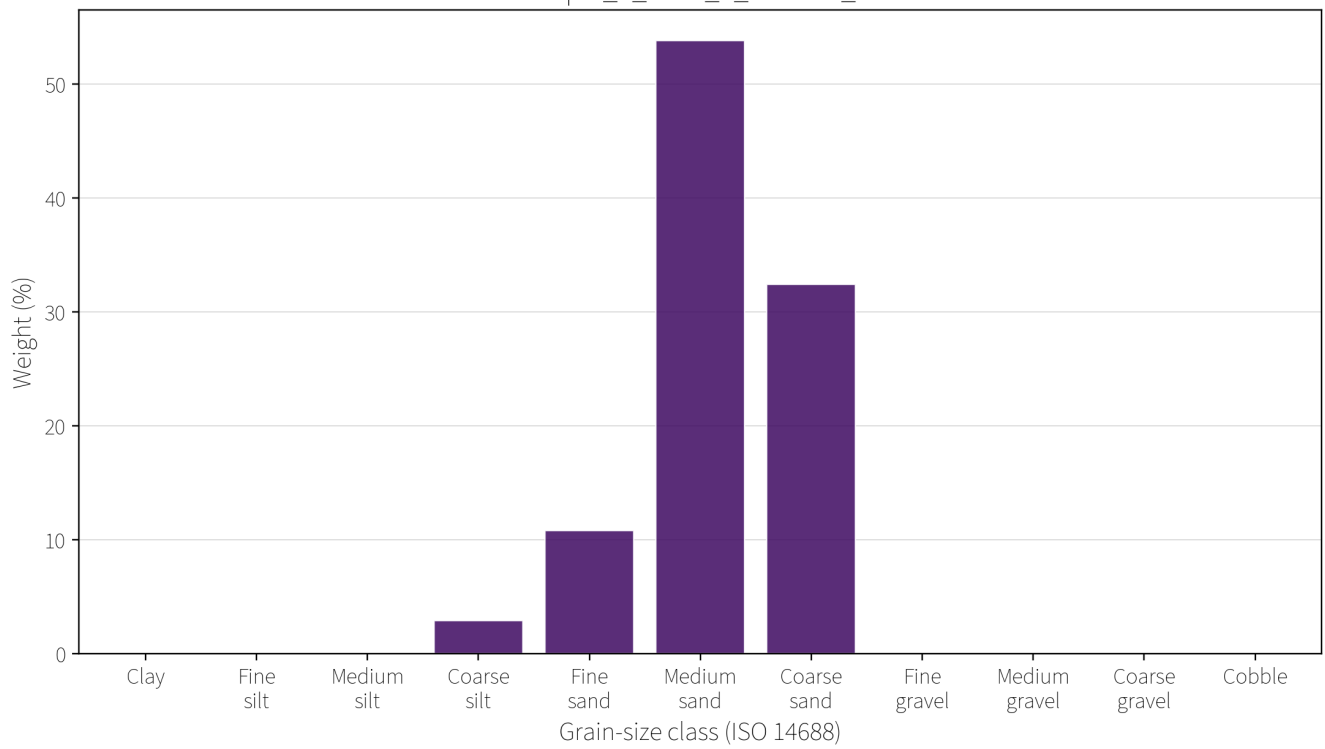


Example_1_Case_2_Vukovic_15C

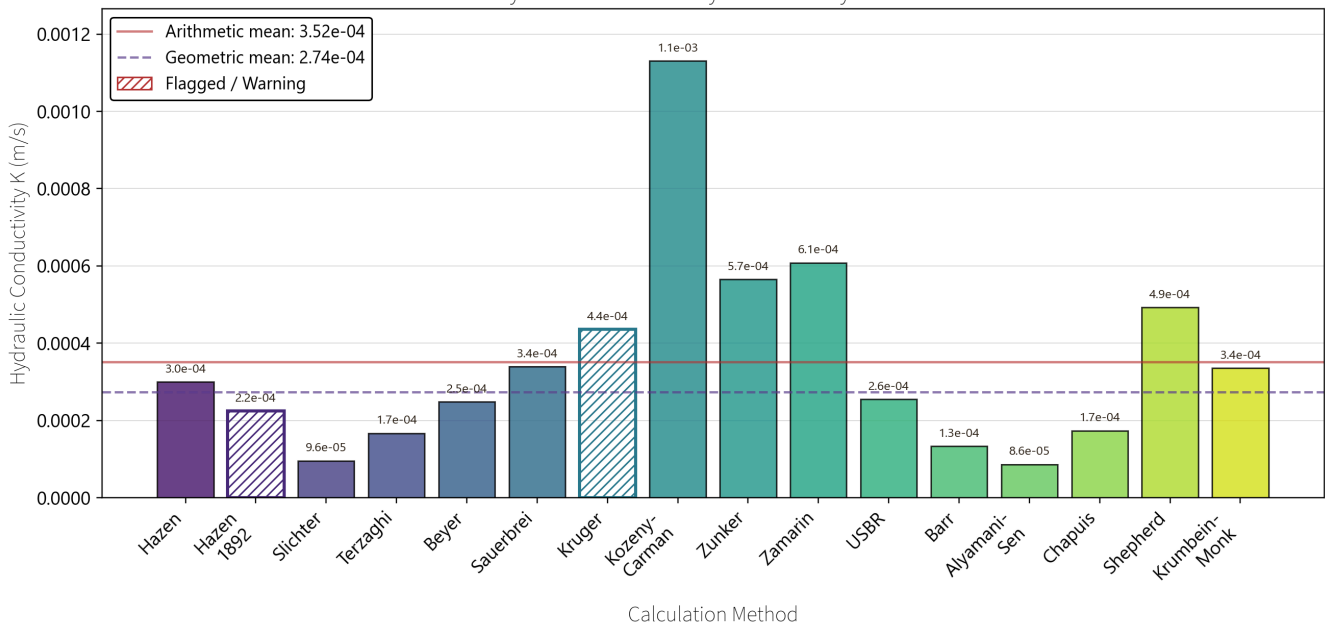
Grain Size Distribution: Example_1_Case_2_Vukovic_15C



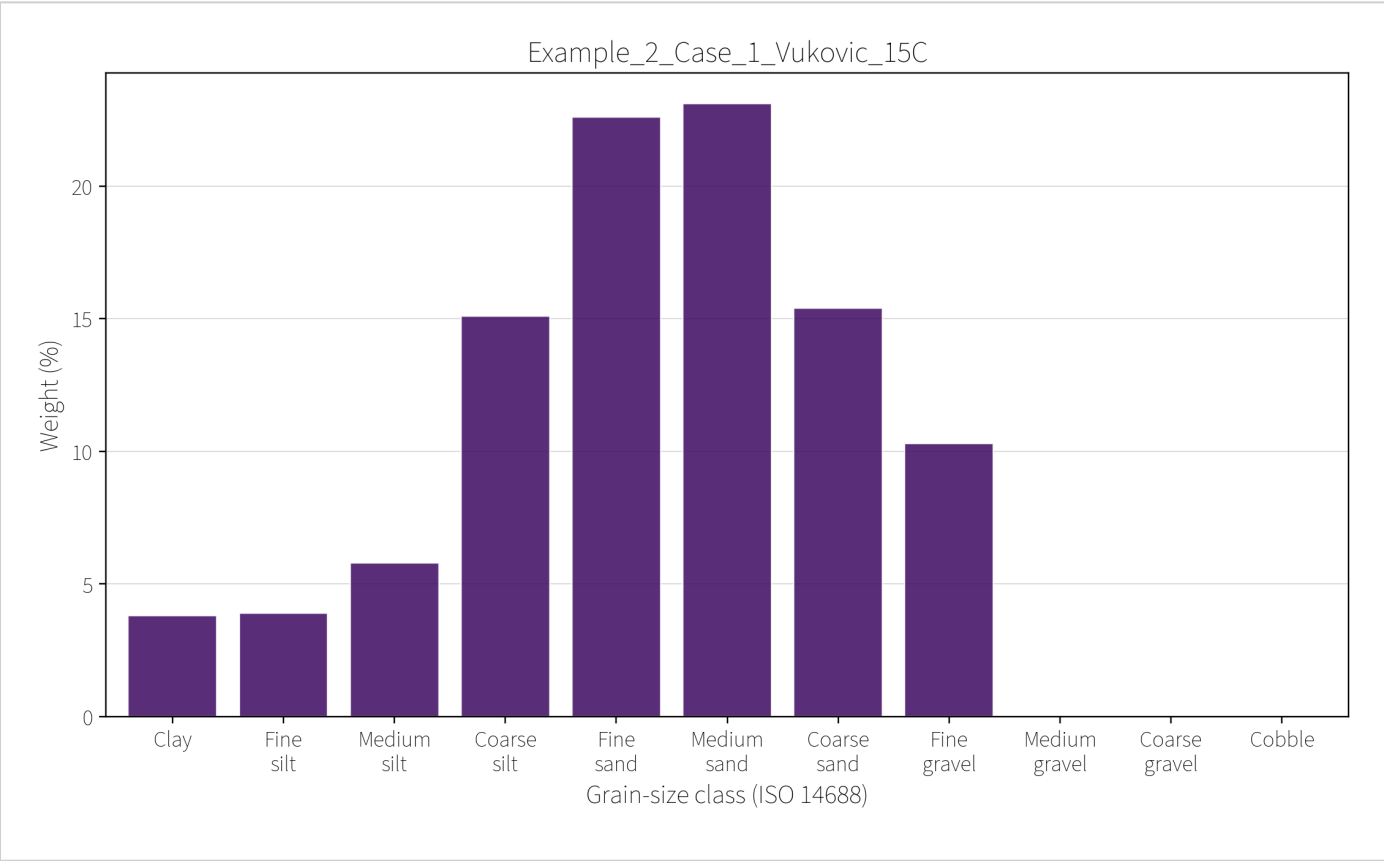
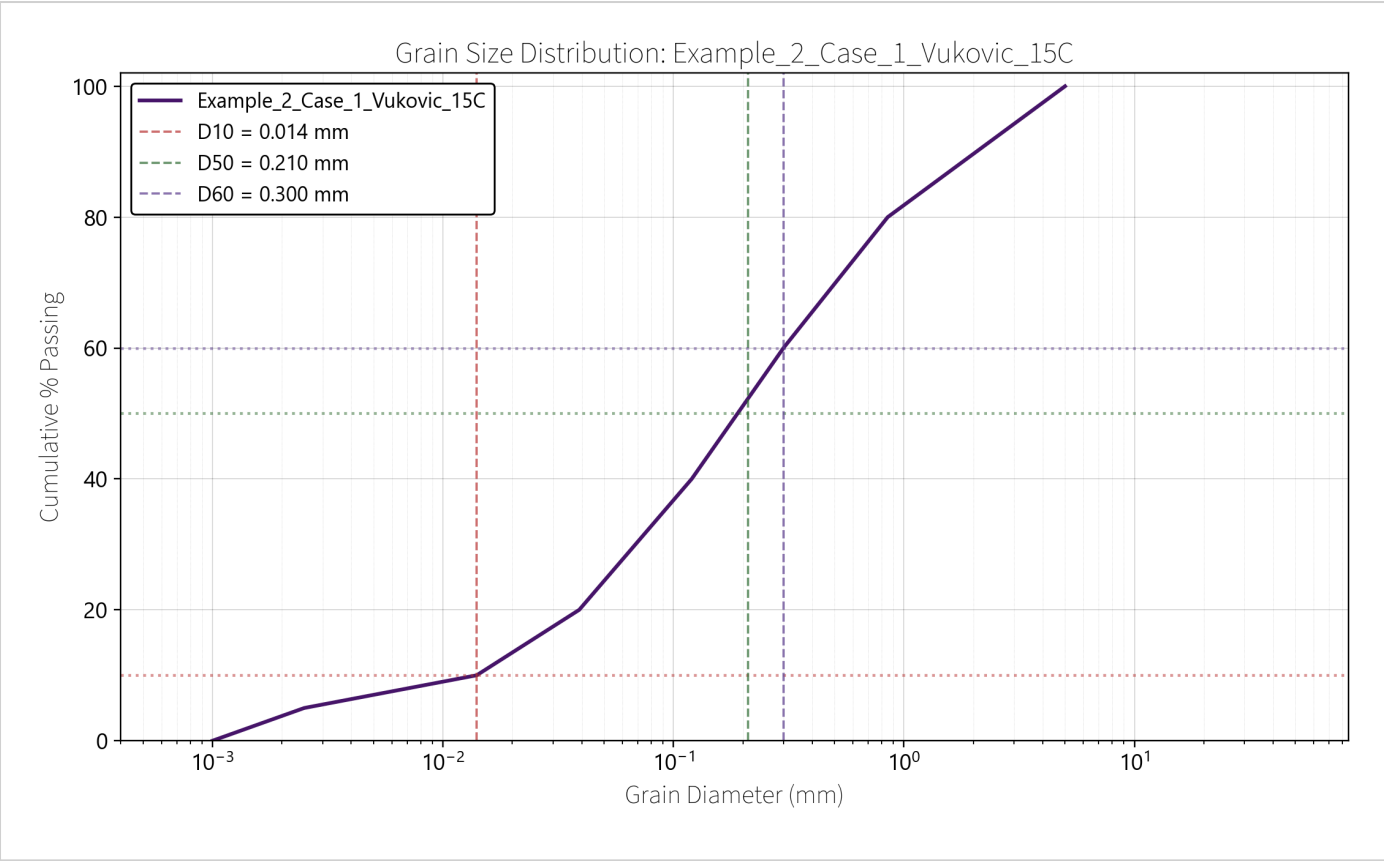
Example_1_Case_2_Vukovic_15C



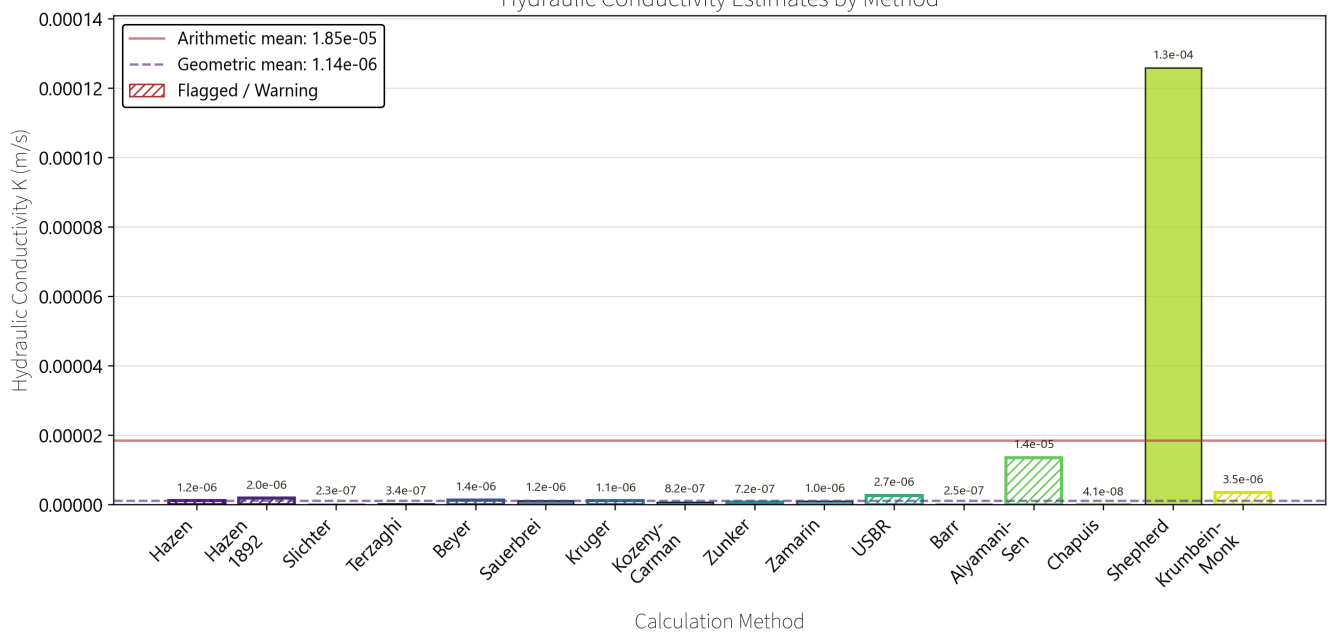
Hydraulic Conductivity Estimates by Method



Example_2_Case_1_Vukovic_15C

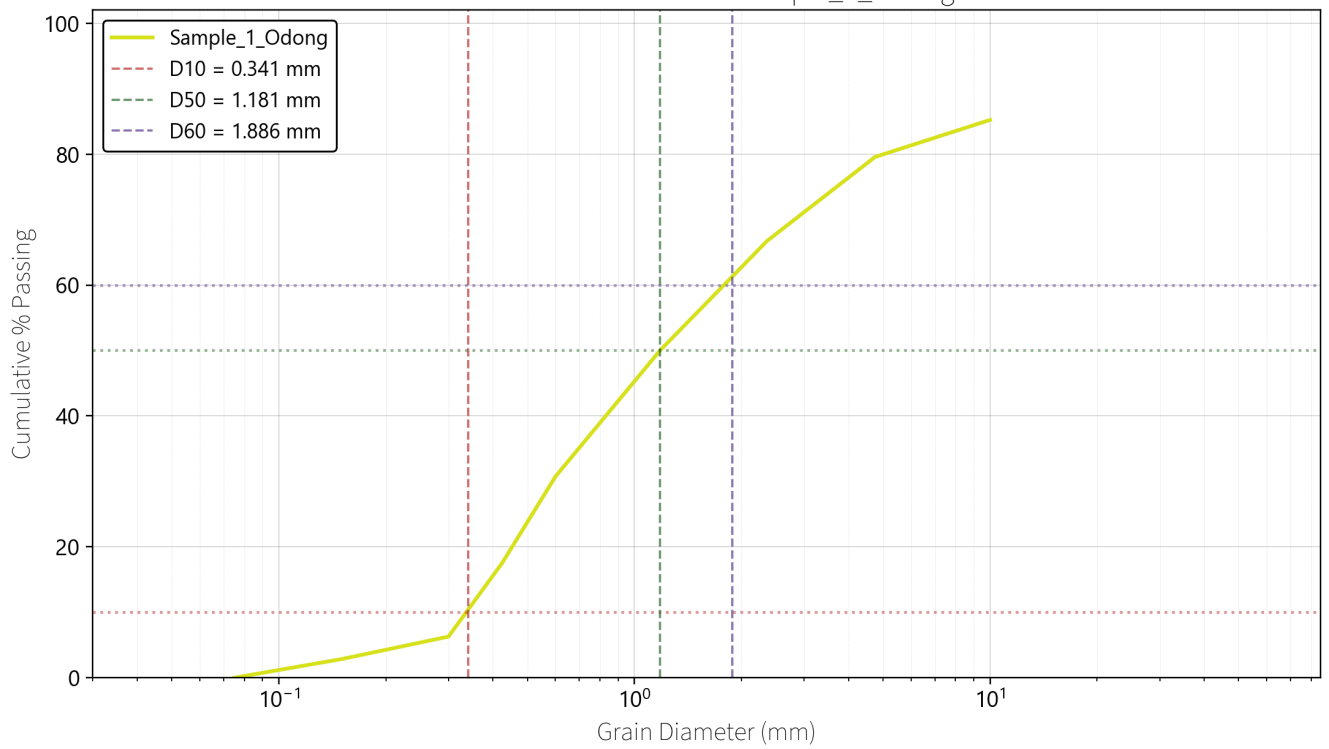


Hydraulic Conductivity Estimates by Method

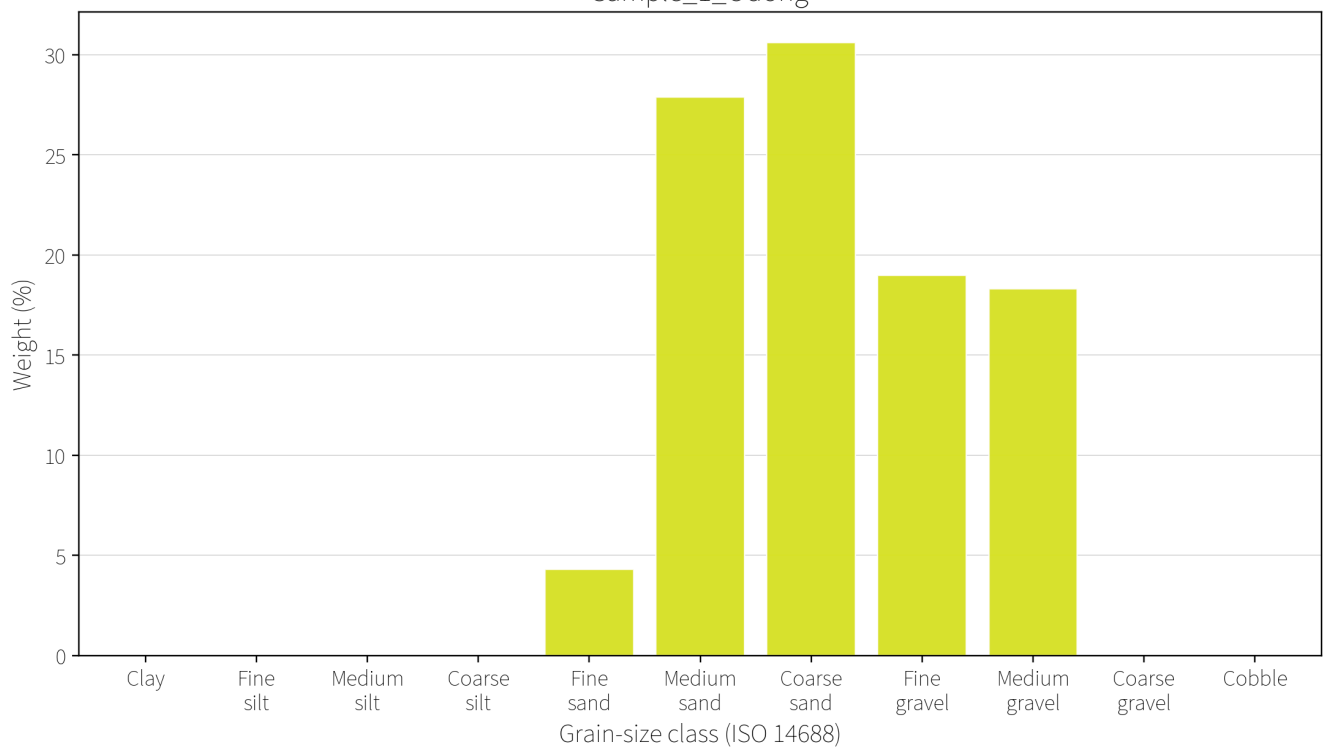


Sample_1_Odong

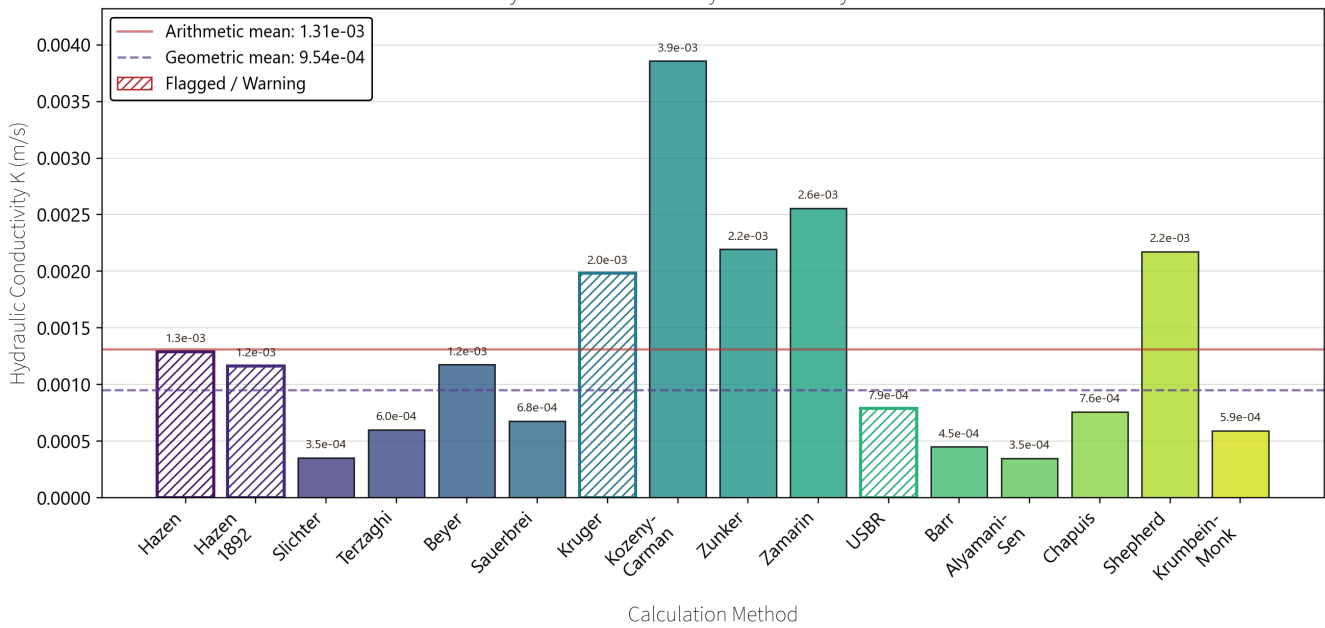
Grain Size Distribution: Sample_1_Odong



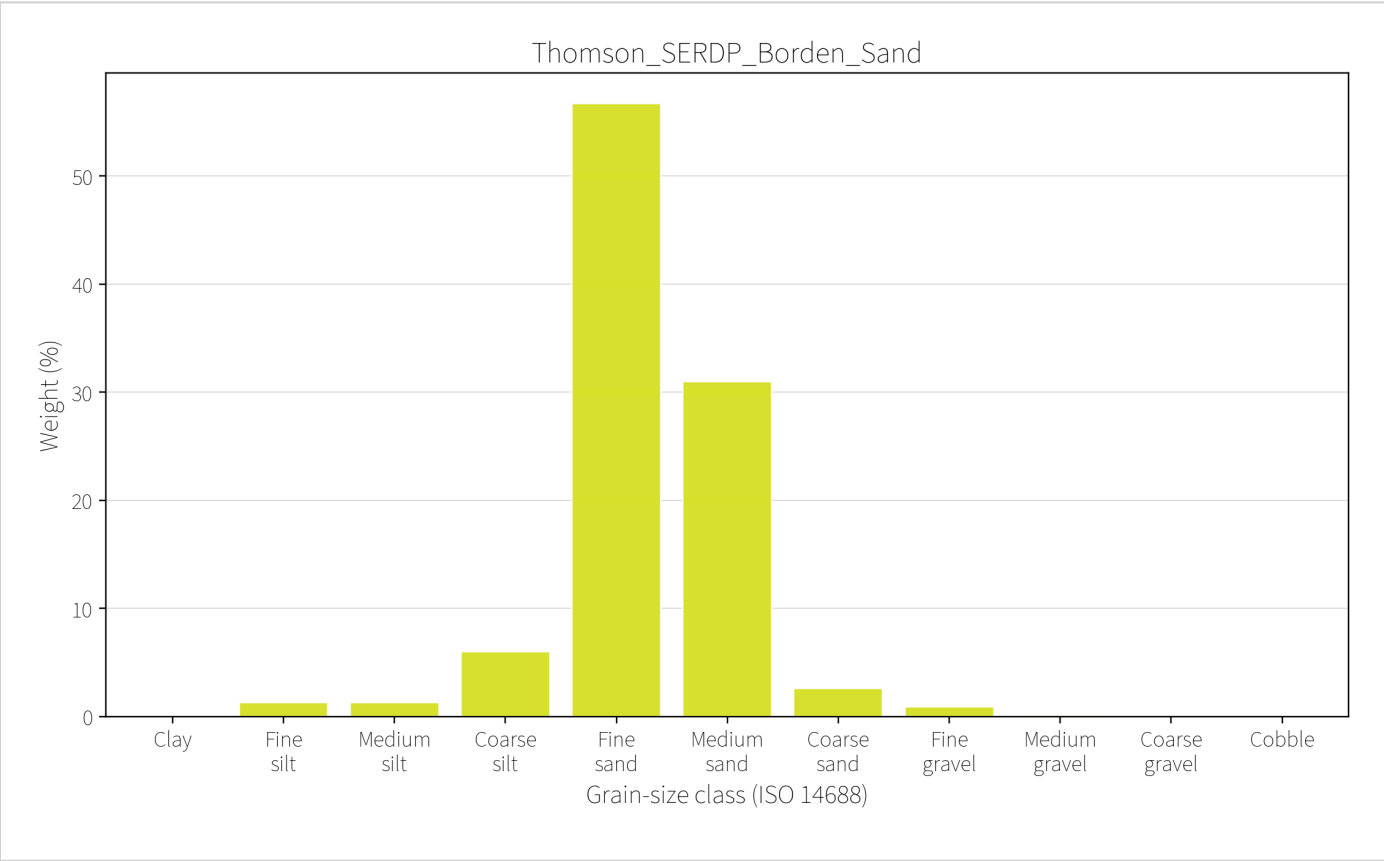
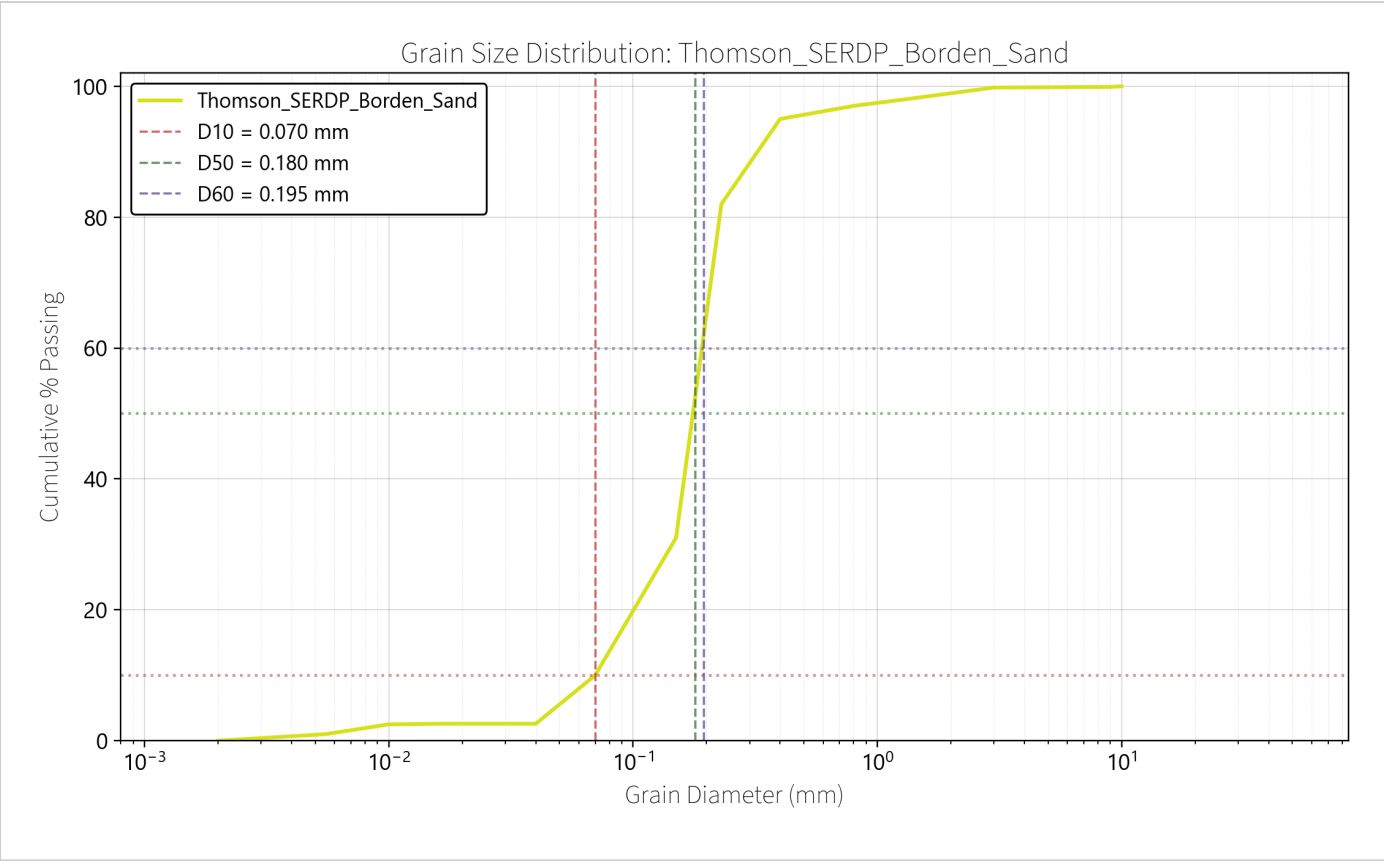
Sample_1_Odong

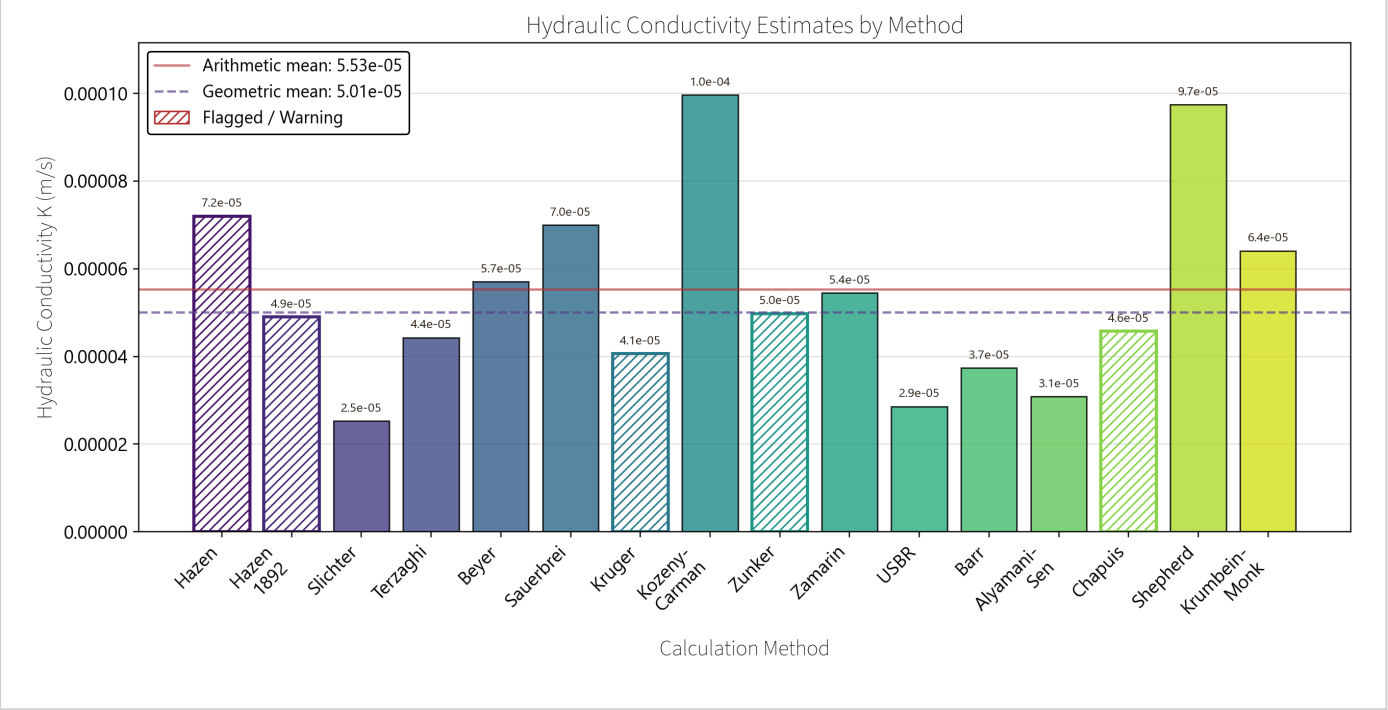


Hydraulic Conductivity Estimates by Method



Thomson_SERDP_Borden_Sand





Interpretation & Discussion

Comparative Analysis

Permeability Characteristics:

- The highest permeability sample is Sample_1_Odong with $K = 9.54 \times 10^{-4}$ m/s, classified as High (Clean Sand).
- The lowest permeability sample is Example_2_Case_1_Vukovic_15C with $K = 1.14 \times 10^{-6}$ m/s, classified as Low (Silt).
- The 837.4-fold difference in permeability reflects the variability in grain size distribution among the samples.

Statistical Overview:

- Average sample K geometric mean: 3.10×10^{-4} m/s
- Standard deviation: 3.41×10^{-4} m/s
- Coefficient of variation: 109.8%